PRE-ALGEBRA (8TH) · UNIT 1: REAL NUMBERS AND EXPONENTS

Space Mission Planner

Use scientific notation, exponents, and roots to plan a crewed mission to Mars.

CP2

2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

NASA has asked your class to help plan the first crewed trip to Mars. The distances are enormous, the radio messages take minutes to arrive, and every module of the habitat has to be sized exactly.

You will use scientific notation, exponent rules, and square and cube roots to plan the trip and design the habitat.



Mission Data

Earth to Moon: 3.84×10^5 km

Earth to Mars (closest): 5.46×10^7 km

Speeds

Rocket: 4×10^4 km per hour

Light: 3×10^5 km per second

Learning Goals

- I can write very large numbers in scientific notation and standard form.
- I can multiply and divide numbers written in scientific notation.
- I can use square roots, cube roots, and exponent rules.

Project Timeline

When	What you will do
Day 1	Big Numbers
Day 2	Travel Time
Day 3	Habitat Design
Final	Mission Plan + reflection

Part 1: Big Numbers

Day 1 · Use the Mission Data Cards.

Mission Data Cards

Moon	Mars (closest)	Mars (farthest)
3.84×10^5 km	5.46×10^7 km	4.01×10^8 km
from Earth	from Earth	from Earth

Rocket Speed	Radio Signal	Food
4×10^4 km	3×10^5 km	2.5×10^3 calories
per hour	per second	per astronaut per day

1. Write the distance from Earth to the Moon, 384,000 km, in scientific notation.

Starter: ___ × 10^___

2. Write 5.46×10^7 km in standard form.

3. About how many times as far is Mars (closest) as the Moon? Divide and round to the nearest whole number.

Part 2: Travel Time

Day 2 · Divide distance by speed to find time.

4. The rocket travels 4×10^4 km per hour. How many hours does it take to reach the Moon?

Starter: *(3.84 ÷ 4) × 10^(5 − 4) = ___*

5. How many hours does the rocket take to reach Mars at its closest? About how many days is that?

6. A radio signal travels 3×10^5 km per second. How many seconds does a message take to reach Mars at its closest?

Part 3: Habitat Design

Day 3 · Use roots and exponent rules to size the habitat.

7. A cube-shaped sleeping module has a volume of 512 cubic meters. How long is each edge?

8. A square solar panel has an area of 49 square meters. How long is each side?

9. Each supply crate holds 2^3 boxes, and each box holds 2^4 meal packs. Write the number of meal packs in one crate as a power of 2 and as a number.

10. Is $\sqrt{50}$ rational or irrational? Between which two whole numbers is it?

Final Product: Mission Plan

Your final product must include:

- Distances in scientific notation and standard form
- Travel times to the Moon and Mars
- Radio delay to Mars
- Habitat sizes found with roots
- A mission poster with one fact written in scientific notation

Reflection

R1. Why is scientific notation useful for space distances?

R2. How do you divide numbers in scientific notation?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Scientific notation computations, exponent rules, and roots are used correctly to plan the mission. Goes beyond the minimum with a creative, realistic choice.	Scientific notation computations, exponent rules, and roots are used correctly to plan the mission.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Mission plan shows distances, times, supplies, and habitat sizes with labeled work and units. Polished, organized, and visually engaging.	Mission plan shows distances, times, supplies, and habitat sizes with labeled work and units. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.